



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2023-24(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

End Semester

Semester: VI

Programme: B. Tech Information Technology

Course: Ethical Hacking

Time: 3 hour

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33603	Course Outcome
CO1	To study the fundamental concepts of ethical hacking, including various phases and types of hackers
CO2	To identify threats, vulnerabilities, and attacks in computer systems and networks
CO3	To explore tools and techniques used in ethical hacking such as foot printing, scanning, social engineering, and password cracking
CO4	To evaluate system and network security measures using penetration testing and vulnerability assessment tools
CO5	Demonstrate awareness of the legal, ethical, and professional issues related to cybersecurity practices

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	What is Google hacking?	CO1	3	2
b.	Define insider threat.	CO2	2	2
c.	What is session hijacking?	CO3	2	2
d.	Define ethical responsibility.	CO4	3	2
e.	What is digital forensics?	CO5	2	2

Group B

Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain types of hackers and their motivations.	CO1	4	5
b.	Describe footprinting techniques in detail.	CO1	5	5
c.	Explain scanning process and its types.	CO1	4	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain types of attackers with real-world scenarios.	CO2	4	5
b.	Analyze application-based attacks with examples.	CO2	5	5
c.	Explain OS vulnerabilities and mitigation techniques.	CO2	4	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain buffer overflow and reverse engineering.	CO3	4	5
b.	Describe session hijacking and email hacking techniques.	CO3	5	5
c.	Explain honeypots and their role in security.	CO3	4	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain cyber laws in India related to hacking.	CO2	4	5
b.	Discuss ethical issues faced by ethical hackers.	CO2	5	5
c.	Explain importance of legal compliance in cybersecurity.	CO2	4	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain malware types: Trojan, worm, virus, rootkit.	CO2	4	5
b.	Describe cybercrime case studies and their impact.	CO2	5	5
c.	Explain steganography and cryptography with differences.	CO2	4	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create